

Technical note: Advancing size-specific dose estimates in CT examinations: Dose estimates at longitudinal positions of scans

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Purpose: American Association of Physicists in Medicine (AAPM) (Report 204) introduced the size-specific dose estimate (SSDE) for the average dose to the center of a fixed-mA scan. International standards establish a method that CT manufacturers and radiation dose index monitoring software may use to calculate SSDE(z) at longitudinal positions of scans with fixed mA or tube current modulation, and its scan range average $\overline{\text{SSDE}(z)}$. We sought to test how accurate SSDE(z) is in tracking the average dose to the transverse slab of an axial image slice (D_{slice}), evaluated with Monte Carlo calculation, in the chest and abdominopelvic examinations.

Methods: We retrospectively identified 65 consecutive adult patients undergoing whole-body CT for transcatheter aortic valve implantation planning. Examination parameters (kV, mA, CTDI_{vol}) were extracted from the DICOM headers. Patient water equivalent diameter $D_W(z)$ was calculated at each image slice, excluding the patient table. A previously validated Monte Carlo simulation (Geant4) program was used to evaluate D_{slice} from the chest and abdominopelvic examinations. Alternatively, SSDE(z) was calculated at each slice. The results of the two methods were compared with descriptive statistical outcomes (R, version 4.0.2).

Results: In chest and abdominopelvic CT examinations, D_{slice} largely changed with anatomic location and uniformly fell off toward scan range edges. Scan range averages $\overline{\text{SSDE}(z)}$ and $\overline{D_{\text{slice}}}$ were consistent within 2.5%–3.1% (median) and 6.3%–10.4% (maximum) in two examinations. On individual image slices, SSDE(z) could be lower or higher than D_{slice} , with deviation ranging from –18.3% to 85% in two edges (2×5 cm) of scan range and from –35.2% to 18.7% in the remaining central region of the scan.

Conclusion: This study provides critical inputs for quality assurance programs. SSDE(z) is useful to track the average dose of all image slices, but further development may be useful for tracking patient dose at the level of individual image slices, especially near a scan range edge.

KEYWORDS

average dose to image slice, comparison, computed tomography, longitudinal position, size-specific dose estimate

Abbreviations: CTDI_{vol}, Volumetric CT dose index; D_{slice} , Average dose to the 2.5 or 5 mm-thick transverse slab of an axial image slice, obtained from Monte Carlo simulation; $\overline{D_{\text{slice}}}$, Scan range average of D_{slice} ; D_W , Water equivalent diameter; SSDE(z), Size-specific dose estimate, at longitudinal position z; $\overline{\text{SSDE}(z)}$, Scan range average of SSDE(z)

1 | INTRODUCTION

With a large number of computed tomographic (CT) scans performed annually (84 million in 2016) in the United States, radiation dose from CT examinations has been closely monitored.^{1–3} CT systems provide volumetric CT dose index (CTDI_{vol}), which does not indicate patient dose. Size-specific dose estimate (SSDE) was introduced for the average dose to the center of a fixed-mA scan range in pediatric and adult CT examinations of the torso or the head.^{4,5} Thereafter, SSDE has been used in various quality assurance programs.⁶ As per AAPM Report 204, SSDE was based on fixed mA and defined for the center of scan range.⁷ Most patient image slices are away from the scan range center, and >67% of CT examinations adopt tube current modulation (TCM), based on an institutional survey of 294 179 CT examinations in 5 years. To move forward, international standards (IEC Publication 62985) establish a method that CT manufacturers and radiation dose index monitoring software may use to calculate SSDE(z) at longitudinal positions of scans and its scan range average $\overline{\text{SSDE}(z)}$, in scan series with fixed mA or TCM.⁸ SSDE(z) may provide an estimate of absorbed dose for the DICOM radiation dose structured report (RDSR) of each image slice. According to the IEC, potential uses of $\overline{\text{SSDE}(z)}$ may include developing diagnostic reference level, dose notification, and dose alert that better consider dose variation within a wide range of patients undergoing a variety of CT examinations.⁸

Radiation dose to the patients from CT examination depends on variables related to patient anatomy and examination techniques. Monte Carlo simulation is the most accurate method for patient dose assessment, although it is a complicated process, requiring patient anatomy modeling, CT simulation software development, and intensive computation, such as on high-performance computing clusters.^{9,10}

The ultimate usefulness of SSDE(z) will depend on how accurate it is in tracking the average dose (D_{slice}) to the transverse slab of individual image slices and the knowledge of D_{slice} and SSDE(z) at longitudinal locations of scans will provide critical input for quality assurance programs. Besides, there is an interest in estimating patient organ doses from CT examinations based on SSDE.^{7,11,12} Especially a regional- or organ-specific estimate of SSDE with TCM may be of interest for improving the accuracy of organ doses.⁷ This study aims to perform Monte Carlo simulation and evaluate D_{slice} in adult body CT examinations, and compare the results with those of SSDE(z) evaluation in the IEC standard.

TABLE 1 Summary of patient demographic data and chest/abdomen/pelvic examination

Parameter		Male	Female
No. of patients (percentage)		37 (56.9%)	28 (43.1%)
Age (years)	Mean	81.1 ± 7.7	82.2 ± 6.5
	Range	64–96	65–92
Weight (kg)	Mean	83 ± 16	71 ± 19
	Range	59–127	43–116
CTDI _{vol} (mGy)	Mean	9.28 ± 2.20	7.55 ± 3.05
	Range	3.71–12.21	2.12–11.38

2 | MATERIALS AND METHODS

2.1 | Voxelized patient cohort

Our Institutional Review Board approved this retrospective study involving anonymized data, with a focus on chest and abdominopelvic examinations. We retrospectively identified 72 consecutive patients undergoing helical CT examinations (at 120 kV) from chest to pelvis in transcatheter aortic valve implantation planning in an ambulatory CT (GE Discovery CT750) room in 2017–2019. Excluding 7 patients with image reconstruction diameters less than patient dimensions, the study cohort included 65 patients. Axial image slice thickness was 5 mm for 8 patients and 2.5 mm for 57 patients, and the total number of images was 986 and 14 244, respectively. In each axial image slice, effective mAs ($= \text{mA} \times t_{\text{rotation}}/\text{pitch}$) were obtained from the DICOM headers, and patient water equivalent diameter D_W was calculated, following AAPM Report No. 220.¹³ Patient table was excluded in the D_W assessment. Table 1 summarizes patient examination data.

2.2 | Calculation of SSDE(z) in the IEC standard

The CT system provided CTDI_{vol} with the scan series average tube current ($\overline{\text{mA}}$). In each axial image slice (z), SSDE(z) was computed,^{4,8}

$$\text{SSDE}(z) = \text{CTDI}_{\text{vol}} \times \text{mA}_z / \overline{\text{mA}} \times a \times \exp(-b \times D_W), \quad (1)$$

where CTDI_{vol} was referenced to the 32-cm CTDI phantom, $a = 3.704369$, $b = 0.03671937 \text{ cm}^{-1}$, and D_W had units of centimeters. For each of chest examination (from the apex to the base of lungs) and abdominopelvic examination (from the lower thorax to pubic symphysis), we also calculated the scan range average of SSDE(z),⁹

$$\overline{\text{SSDE}}(z) = \sum_{i=1}^N \frac{1}{N} \times \text{SSDE}(z_i), \quad (2)$$

where N was the number of axial image slices of scans.

2.3 | Monte Carlo calculation of D_{slice} in chest and abdominopelvic examinations

To assess patient doses from the chest and abdominopelvic examinations, we used a previously validated C++ program for Monte Carlo calculations, based on Geant4 (version 9.1).¹⁰ The program explicitly simulated the geometry of a Siemens Definition CT system. The CT scan modeling included large focal spot size, inherent beam filtrations, body bowtie filter, beam collimation (19.2 mm), penumbra, as well as low energy photon and electron processes for the photoelectric effect, Compton scattering, and Rayleigh scattering with energies above 1 keV. The material and dimension of the bowtie filter were provided by the CT manufacturer under a nondisclosure agreement. Gantry rotation was simulated with 360 projections per full rotation. The simulation program was validated with ion chamber-based CTDI₁₀₀ measurement on the CT system. For each patient, all available (N_{slice}) axial images (from chest to pelvis) were imported to construct a voxelized phantom of $512 \times 512 \times N_{\text{slice}}$ voxels. The Hounsfield unit (HU) numbers were converted to materials based on a previous method.¹⁴ Air was used for $\text{HU} \leq -1000$. The higher HU values were converted to 25 elemental compositions and 253 mass densities. The patient table was included in the DICOM images and voxelized phantom for simulation but was excluded in dose calculation. The scanning technique was at 120 kV, beamwidth = 19.2 mm, and helical pitch = 1. Helical CT scanning was simulated with 360 projections and 360 patient table translations per gantry full rotation, each translation by (helical pitch) $\times$ (beam width)/360. For simulating TCM scans, energy depositions were weighted by the mA_z to $\overline{\text{mA}}$ ratio, where mA_z was the tube current at the slice irradiated by the primary beam and $\overline{\text{mA}}$ was the scan range average tube current. Effective $\overline{\text{mAs}}$ ($= \overline{\text{mA}} \times t_{\text{rotation}}/\text{pitch}$) was used to calculate CTDI_{vol}, based on the average mA , rotation time, pitch, and CTDI_{vol} of the original patient examination. In this work, each chest or abdominopelvic examination was simulated with a sample of $\geq 1.2 \times 10^9$ photon histories for statistical error $< 1\%$. The average dose to the transverse slab of an axial image slice was computed with the total energy deposited in the slice divided by the total mass of all voxels of the slice, excluding the patient table. The result was normalized to the CTDI_{vol} of the chest or abdominopelvic examination, respectively.^{15,16}

The patient examination was limited to the anatomy region of clinical indication. The images of 65 patients contained 2 (0–3.75) cm [median(range)] superior to the apex of lungs and 3.25 (0.5–9) cm inferior to the pubic symphysis. In the simulation, patient anatomic modeling based on these images did not contain full patient anatomy, and the dose near the above edges might be smaller than that of patient examination. To quantify the backscatter loss, we identified another patient (male; age, 61 years; height, 173 cm; weight, 75 kg) undergoing PET/CT attenuation correction scan (from skull base to middle thighs) on Siemens Biograph 64. A total of 198 axial images (slice thickness, 5 mm) were imported for two helical CT scans, with (a) full patient anatomy; (b) truncated patient anatomy, in which all slices superior to the apex of lungs or inferior to pubic symphysis were set to be air. Each scan was from the apex of lungs to pubic symphysis, at 120 kV, fixed mA , beamwidth = 19.2 mm, and helical pitch = 1. Each simulation used 1.6×10^{10} photon histories, and total energy deposition in each image slice was calculated, excluding the patient table. The truncated to full anatomy ratio of energy deposition was used for backscatter loss correction of the above 65 chest examinations, based on their image slice distance to the chest superior edge slice. A similar correction was performed for the above 65 abdominopelvic examinations, based on their image slice distance to the pelvic inferior edge slice.

2.4 | Statistical analysis

Patient age, weight, and radiation output (CTDI_{vol}) from CT examinations were analyzed by their means and ranges. The study cohort included image reconstruction slice thickness of 5 mm for 8 patients and 2.5 mm for 57 patients. SSDE(z) was compared with D_{slice} from the Monte Carlo simulation. The difference in magnitude between them was assessed in five percentiles (10th, 25th, 50th, 75th, and 95th) for each individual patient, for the group with a slice thickness of 5 mm, and for three weight groups (small, normal, large) with a slice thickness of 2.5 mm, in each of chest and abdominopelvic examinations. The difference was separately evaluated in two groups, two edges (2×5 cm) of scan range and its remaining central region. Data analysis was performed with statistical software (R, version 4.0.2; The R Project for Statistical Computing, <https://r-project.org>).

3 | RESULTS

For a male (age, 61 years; height, 173 cm; weight, 75 kg) undergoing chest/abdomen/pelvic examination from the apex of lungs to pubic symphysis (at 120 kV, fixed mA , and CTDI_{vol} = 10 mGy), Figure 1 (middle panel) shows

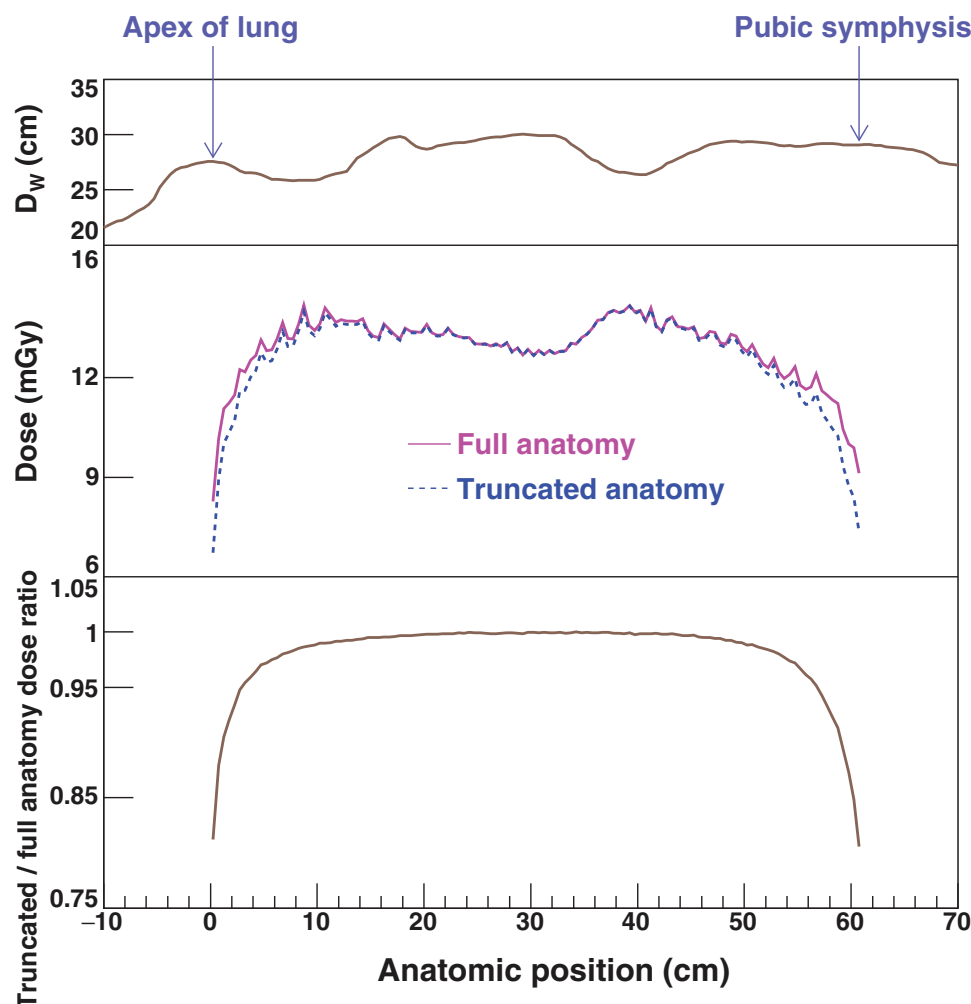


FIGURE 1 Graphs show water equivalent diameter (D_w), radiation dose from Monte Carlo simulation, and dose ratio for a male (age, 61 years; height, 173 cm; weight, 75 kg) undergoing CT examinations from the apex of lungs to pubic symphysis, at 120 kV, fixed mA, and $CTDI_{vol} = 10$ mGy. While full patient anatomy was from skull base to middle thighs, the truncated anatomy was from the apex of lungs to pubic symphysis

dose profiles from two patient anatomy settings in Monte Carlo simulation, with (a) full anatomy (from skull base to middle thighs) and (b) the truncated anatomy (from the apex of lungs to pubic symphysis). While the patient D_w profile is shown in Figure 1 (top panel), this figure (bottom panel) also shows the truncated to full anatomy dose ratio, which was essentially unity at anatomic locations that were >10 cm away from both truncated edges. However, the ratio decreased as the anatomic location approached either edge, being as low as 0.81.

Figure 2 shows the profiles of D_w , mA, radiation doses from Monte Carlo simulation and SSDE(z) estimation, and their difference in magnitude for three patients undergoing chest and abdominopelvic examinations. With a low weight female (age, 75 years; height, 163 cm; weight, 48 kg), the mA profile had a flat segment (length, 21.5 cm) of 75 mA, due to a setting of minimum mA in the original protocol. Figure 2a shows

that D_{slice} largely changed with anatomic position and fell off toward either end of the scan range. SSDE(z) also changed with anatomic position but did not fall off toward the end of the scan range. The difference between SSDE(z) and D_{slice} was $[2.1 \pm 11.2 \text{ } (-11.5, 38.3)]\%$ [mean(minimum, maximum)] in two edges of chest examination, $[-16.1 \pm 4.2 \text{ } (-24.7, -5.3)]\%$ in the remaining central region of the examination; $[-2.9 \pm 11 \text{ } (-18.3, 26.7)]\%$ in two edges of abdominopelvic examination, and $[-7.6 \pm 3.5 \text{ } (-13.6, -1.5)]\%$ in the remaining central region of the examination. For a normal weight male (age, 91 years; height, 170 cm; weight, 70 kg), Figure 2b shows that mA largely changed with the anatomic position in the TCM-enabled examination, and dose also largely changed with anatomic position. The difference between SSDE(z) and D_{slice} was $[8.5 \pm 13.3 \text{ } (-13.6, 50.7)]\%$ in two edges of chest examination, $[-12 \pm 9 \text{ } (-28, 1.6)]\%$ in the remaining central region of the examination; $[3.9 \pm 11.7 \text{ } (-6.4, 51.9)]\%$ in

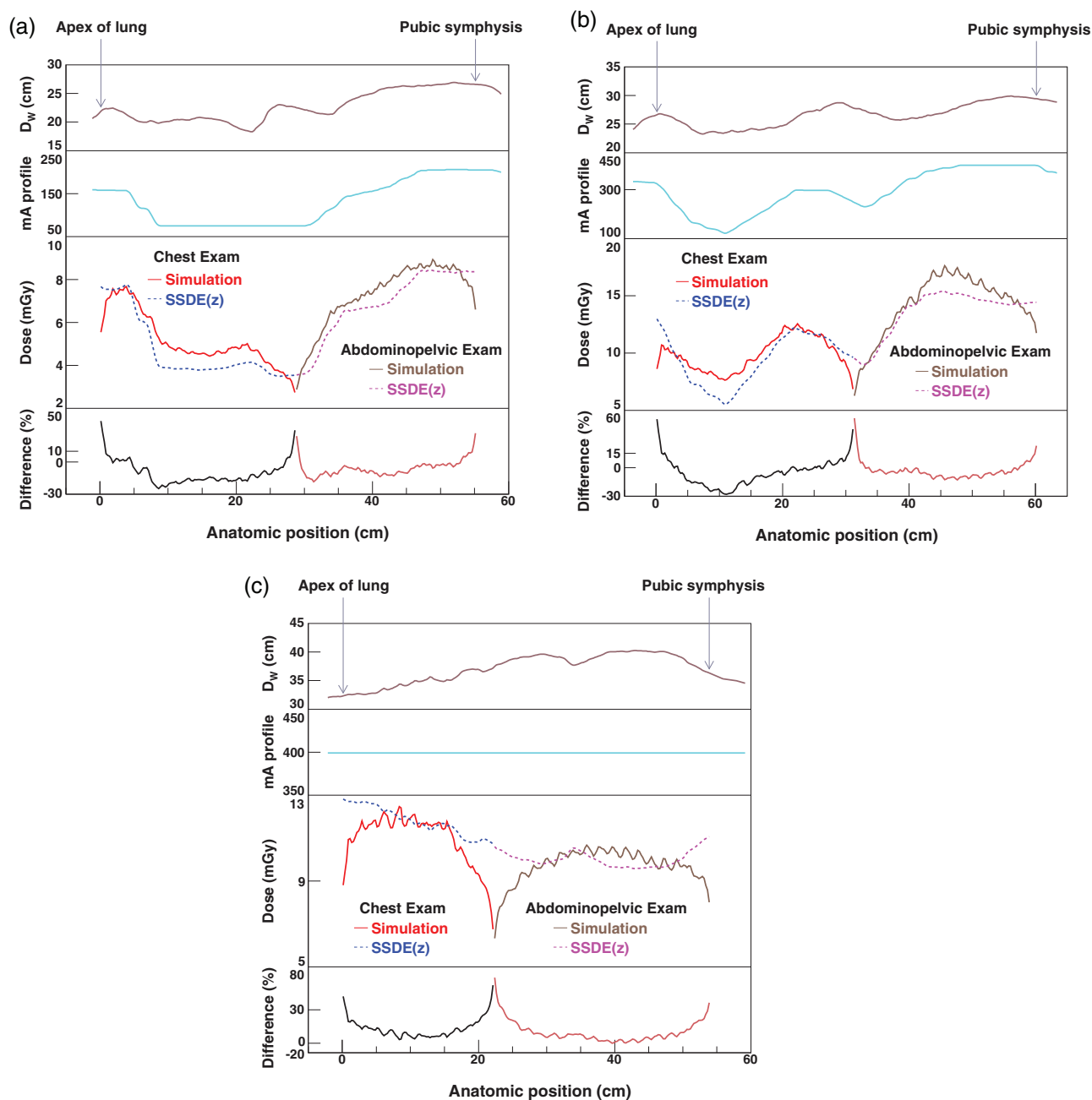


FIGURE 2 Graphs show profiles of water equivalent diameter (D_w), mA, radiation dose from chest and abdominopelvic examinations (at 120 kV, beam width = 19.2 mm, helical pitch = 1), and dose difference between SSDE(z) and Monte Carlo simulation. (a) Small weight female (age, 75 years; height, 163 cm; weight, 48 kg). (b) Normal weight male (age, 91 years; height, 170 cm; weight, 70 kg). (c) Obese female (age, 65 years; height, 157 cm; weight, 108 kg).

two edges of abdominopelvic examination, and $[-7 \pm 3.2$ ($-12.8, -0.3$)]% in the remaining central region of the examination. For an obese female (age, 65 years; height, 157 cm; weight, 108 kg), tube current was at 399 mA, due to a setting of maximum mA in the original protocol. Figure 2c shows that while D_{slice} largely changed with anatomic position, SSDE(z) varied inversely with D_w . The difference between SSDE(z) and D_{slice} was $[17.4 \pm 12.1$ (4.6, 58.7)]% in two edges of

chest examination, $[1.4 \pm 3.5$ ($-4.7, 9$)]% in the remaining central region of the examination; $[17.7 \pm 14$ ($-1.2, 67.6$)]% in two edges of abdominopelvic examination, and $[-2.1 \pm 3.7$ ($-8.8, 8.5$)]% in the remaining central region of the examination.

The difference in magnitude between SSDE(z) and D_{slice} was further examined in five percentiles. Figure 3a shows the results of eight patients (from the first eight examinations by date), whose image slice thickness was

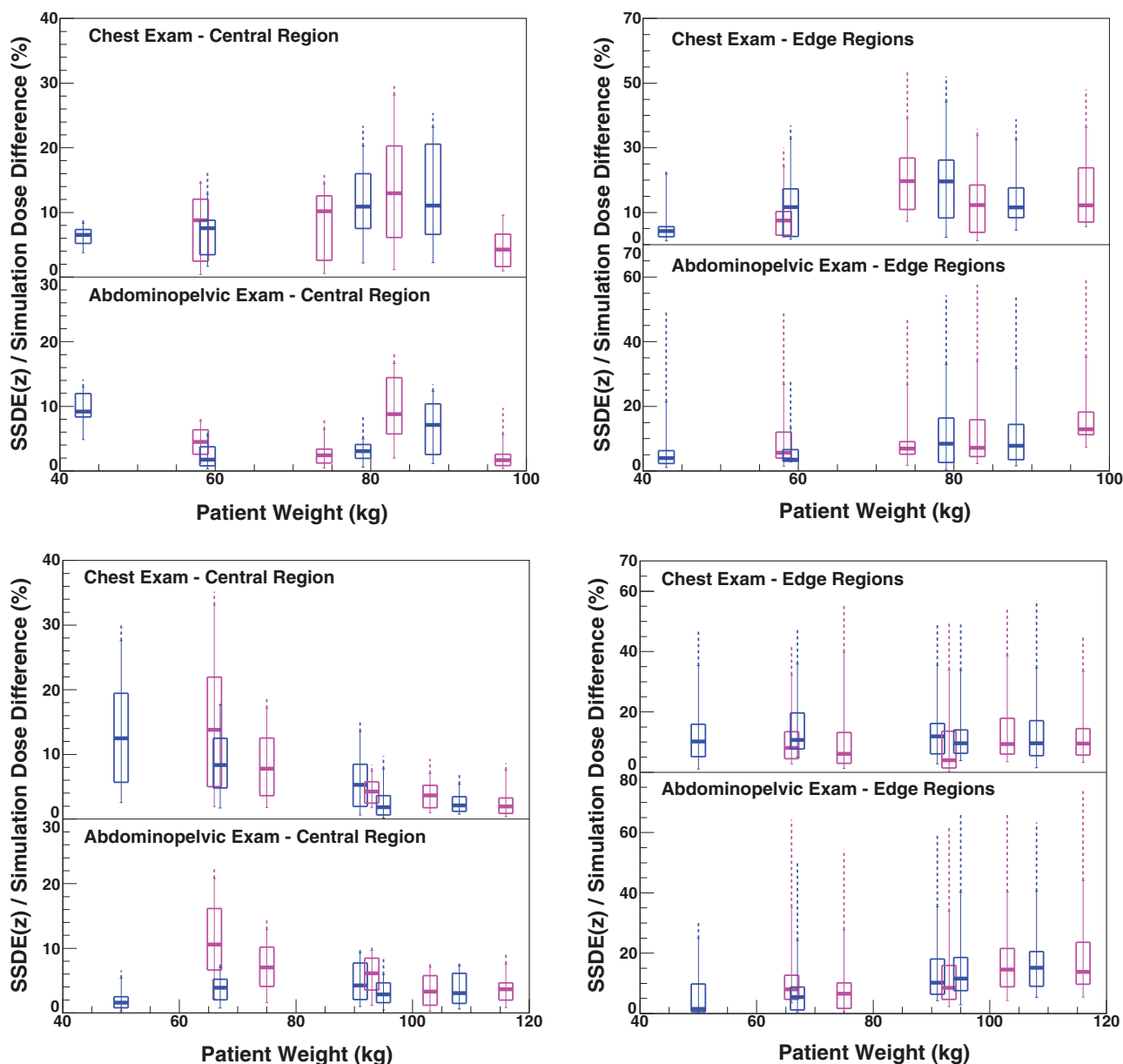


FIGURE 3 Graphs show five percentiles (10th, 25th, 50th, 75th, 95th) and the maximum of dose difference in magnitude between SSDE(z) and Monte Carlo simulation. Results are shown for chest and abdominopelvic examinations. Alternative color in weight sequence is displayed to distinguish patients. Top row: Eight patients (from first eight examinations by date), whose image slice thickness was 5 mm. Top Left: central scan region, excluding two edges (2×5 cm) of scan range. Top Right: two edges (2×5 cm) of scan range. Bottom row: Ten patients (from last 10 examinations by date), whose image slice thickness was 2.5 mm. Bottom Left: central scan region, excluding two edges (2×5 cm) of scan range. Bottom Right: two edges (2×5 cm) of scan range

5 mm. With a weight range of 43–97 kg, the median or maximum difference was 4.3%–21.2% or 22.6%–54.1% for two edges (2×5 cm) of chest examination, 4.5%–14.2%, or 8.9%–29.6% for the remaining central region of the examination; 3.7%–13% or 27.8%–58.9% for two edges of abdominopelvic examination, and 1.7%–9.3% or 6%–18.3% for the remaining central region of the examination, respectively. Figure 3b shows the results of ten patients (from the last 10 examinations by date), whose image slice thickness was 2.5 mm.

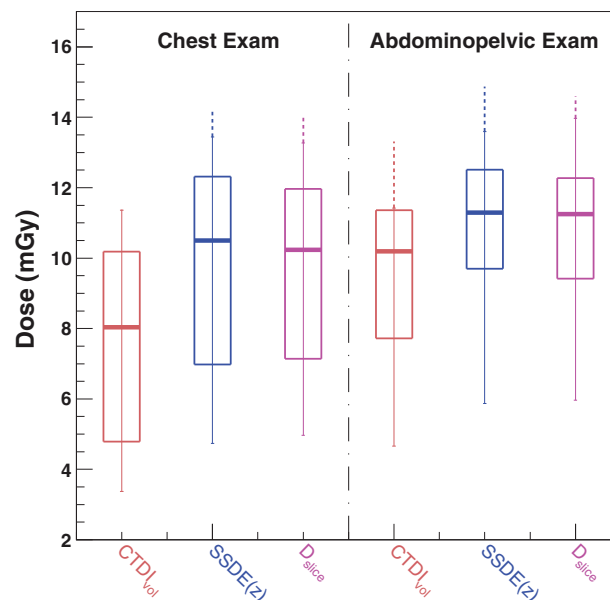
With a weight range of 50–116 kg, the median or maximum difference was 4.1%–11.9% or 42.3%–57.1% for two edges of chest examination, 1.9%–14.1% or 6.9%–35.2% for the remaining central region of the examination; 1.9%–15.3% or 30.3%–73.8% for two edges of abdominopelvic examination, and 1.6%–10.7% or 6.6%–22.5% for the remaining central region of the examination, respectively.

The difference in magnitude between SSDE(z) and D_{slice} is additionally examined in Table 2. For eight

TABLE 2 Difference (10th, 25th, 50th, 75th, 95th percentiles, and maximum) in magnitude between SSDE(z) and D_{slice} from Monte Carlo simulation

CT Exam	T (mm)	W (kg)	M	N	Difference (%) in magnitude in central scan region, excluding two edges (2 × 5 cm)					Difference (%) in magnitude in two edges (2 × 5 cm) of scan range						
					10th	25th	50th	75th	95th	Max	10th	25th	50th	75th	95th	Max
Chest	5	43–97	8	391	1.2	4.2	8.2	12.8	21.1	29.6	2.4	5.1	11.1	21.9	37	54.1
	2.5	48–69	18	1844	2.4	6.2	12.4	18.9	25.6	35.2	2.3	5	10.1	18.2	40.7	85
		70–89	25	2583	1.1	2.6	6.4	13	23.5	35.1	3	6.1	12.2	20.5	41.6	69.7
Abdomen and pelvis	5	90–127	14	1459	0.5	1.4	3	5.4	19.4	26	4	7	11.1	19	38.6	73
		43–97	8	493	0.7	1.8	4	8.3	13.5	18.3	1.7	3.8	7.2	12.9	33.9	58.9
		48–69	18	2115	1.3	3.2	6.2	10.1	14.9	22.5	1.1	3	7.1	13.1	32.2	74.1
	2.5	70–89	25	3185	0.9	2.2	4.7	7.2	11.4	16.2	2	4.7	9.9	17.3	40.7	71.8
		90–127	14	1885	0.7	1.7	3.7	6.1	8.7	17.6	4.6	7.8	13.8	21.1	44.1	73.8

Abbreviations: M, no. of patients; N, no. of image slices; T, slice thickness; W, weight.

**FIGURE 4** Graph shows scan range mean values of $CTDI_{\text{vol}}$, $SSDE(z)$, and D_{slice} of 65 adults in chest and abdominopelvic examinations. Each box and whisker plot shows five percentiles (10th, 25th, 50th, 75th, 95th) and the maximum among 65 patients

patients with 5-mm slice thickness, the median or maximum difference was 11.1% or 54.1% for two edges (2×5 cm) of chest examination, 8.2% or 29.6% for the remaining central region of the examination; 7.2% or 58.9% for two edges of abdominopelvic examination, and 4% or 18.3% for the remaining central region of the examination, respectively. For 57 patients with 2.5-mm slice thickness, the median difference was 10.1%, 12.2%, and 11.1% for three weight ranges (48–69, 70–89, 90–127 kg) for two edges of chest examination, 12.4%, 6.4%, and 3.0% for these weight ranges for the remaining central region of the examination; 7.1%, 9.9%, and 13.8% for these weight ranges for two edges of abdominopelvic examination, and 6.2%, 4.7%, and 3.7% for these weight ranges for the remaining central region of the examination. The maximum difference was 85%, 69.7%, and 73% for these weight ranges for two edges of chest examination, 35.2%, 35.1%, and 26% for these weight ranges for the remaining central region of the examination; 74.1%, 71.8%, and 73.8% for these weight ranges for two edges of abdominopelvic examination, and 22.5%, 16.2%, and 17.6% for these weight ranges for the remaining central region of the examination.

For each patient, we calculated arithmetic mean values ($CTDI_{\text{vol}}$, $SSDE(z)$, D_{slice}) over the scan range of the chest or abdominopelvic examination, respectively. Figure 4 shows the results of five (10th, 25th, 50th, 75th, 95th) percentiles and the maximum of $CTDI_{\text{vol}}$, $SSDE(z)$, and D_{slice} in a cohort of 65 patients. For the chest examinations, the median and maximum values were 8.0 and 11.4 mGy ($CTDI_{\text{vol}}$); 10.5 and 14.2 mGy ($SSDE(z)$); 10.2

and 14.0 mGy ($\overline{D_{\text{slice}}}$). For the abdominopelvic examinations, the median and maximum values were 10.2 and 13.3 mGy (CTDI_{vol}); 11.3 and 14.9 mGy ($\text{SSDE}(z)$); 11.2 and 14.6 mGy ($\overline{D_{\text{slice}}}$). The five (10th, 25th, 50th, 75th, 95th) percentiles and maximum difference in magnitude between $\text{SSDE}(z)$ and $\overline{D_{\text{slice}}}$ were, respectively, 0.3%, 0.8%, 3.1%, 4.8%, 7.5%, 10.4% for chest examinations, and 0.3%, 1.5%, 2.5%, 3.5%, 5.8%, 6.3% for abdominopelvic examinations. Individual patient demographic information, three-dose descriptors, and the difference between $\text{SSDE}(z)$ and $\overline{D_{\text{slice}}}$ can be found in Table S1.

4 | DISCUSSION

Our study revealed the complexity of $\overline{D_{\text{slice}}}$ in CT examinations. It changes with local tube current and patient attenuation but is largely affected by mA line-shape and D_W profile, due to the strong influence of scatter from other locations.^{17,18} A substantial decrease of such scatter near scan range edges leads to $\overline{D_{\text{slice}}}$ uniformly falling off, irrespective of mA line-shape and D_W profile. Moreover, the $\overline{D_{\text{slice}}}$ profile followed distinct line-shapes for the patients with a small weight, normal weight, or large weight in chest examinations (Figure 2), because of unique patient habitus and specific mA line-shapes in such examinations. $\overline{D_{\text{slice}}}$ near the central sections of lungs might be higher or lower than that at the upper or lower thoracic region, depending on specific patient examination.

The SSDE in AAPM Report 204 was originally intended to provide an estimate of the patient dose at the scan range center of fixed-mA scans of 15–30 cm (scan length).⁴ The IEC has established a method that CT manufacturers and dose tracking software may use to calculate $\text{SSDE}(z)$ at longitudinal positions of scans and its scan range average $\overline{\text{SSDE}(z)}$ for fixed mA or TCM.⁸ In this study with 65 adult patients, $\overline{\text{SSDE}(z)}$ and $\overline{D_{\text{slice}}}$ were consistent within 3.1% (median) and 10.4% (maximum) in chest examinations, and 2.5% (median) and 6.3% (maximum) in abdominopelvic examinations. But on individual image slices, $\text{SSDE}(z)$ could be substantially lower or higher than $\overline{D_{\text{slice}}}$, with deviation ranging from –13.6% to 85% in two edges (2×5 cm) of chest examination, from –35.2% to 18.7% in the remaining central region of the examination; from –18.3% to 74.1% in two edges of abdominopelvic examination, and from –22.5% to 11% in the central region of the examination. Especially when the mA curve has a dip (local minimum), $\text{SSDE}(z)$ might be substantially lower than $\overline{D_{\text{slice}}}$, see Figure 2b. Near scan range edges, $\text{SSDE}(z)$ was consistently higher than $\overline{D_{\text{slice}}}$ by 3.5%–85%. These findings provide critical inputs for patient dose tracking program, patient organ dose evaluation, and fetal dose

assessment for pregnant patients, using $\overline{D_{\text{slice}}}$, $\text{SSDE}(z)$, and $\overline{\text{SSDE}(z)}$ evaluated in the IEC standard.

4.1 | Limitations

This study included 65 patients undergoing CT examinations on GE Discovery CT750, but whose beam width (40 mm) and beam filters were not used in the Geant4 simulation. The patient table was included in the radiation transport calculation, but it was excluded in the D_W assessment. Our focus was on adult body examinations with long scan lengths (20–33 or 25–39 cm in the chest or abdominopelvic examinations, respectively). Scan lengths for body CT span a large range, from subcentimeter to about 1 m. As cumulative dose from CT examinations largely changes with scan length,^{19,20} future studies may examine $\overline{D_{\text{slice}}}$, perhaps in comparison with $\text{SSDE}(z)$, in short and intermediate scan lengths. CT examinations in the head/neck region and pediatric patients may also be explored in the future.

SSDE in AAPM Report 204⁴ is applicable only to a single series. But in the institutional survey, 51.3% of CT examinations involved ≥ 2 acquisition series. At the same longitudinal location, $\overline{D_{\text{slice}}}$ or $\text{SSDE}(z)$ may be accumulated over multiple series. This may achieve exam-level dose monitoring²¹ and may be explored in the future.

5 | SUMMARY

The IEC (Publication 62985) adopted $\text{SSDE}(z)$ for the DICOM RDSR of all axial image slices, acquired with fixed mA or TCM. We showcase chest and abdominopelvic CT, in which $\overline{\text{SSDE}(z)}$ can track the average dose of all image slices of scans (median difference, 2.5%–3.1%). On individual image slices, $\text{SSDE}(z)$ could be lower or higher than $\overline{D_{\text{slice}}}$, with deviation ranging from –18.3% to 85% in two edges (2×5 cm) of scan range, and from –35.2% to 18.7% in the remaining central region of the scan, due to the strong influence of scattered radiation from other locations. These findings provide critical inputs for CT quality assurance programs. Further development may be needed for tracking patient dose at the level of individual image slices, especially near a scan range edge. Additional development may be useful for multiscan examinations, which is currently excluded in the US diagnostic reference levels and achievable doses.³

DATA AVAILABILITY STATEMENT

The summary results that support the findings of this study are available as supplementary material associated with the online journal article.

CONFLICT OF INTEREST

The authors have no conflicts to disclose.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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